

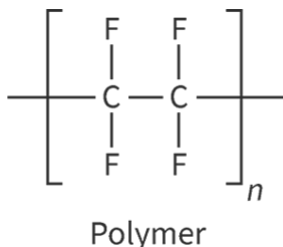
Addition polymers

Learning outcomes

By the end of this section you should be able to:

- Represent the repeating unit of an addition polymer from given monomer structures.
- Deduce the monomer for an addition polymer given the polymer structure.

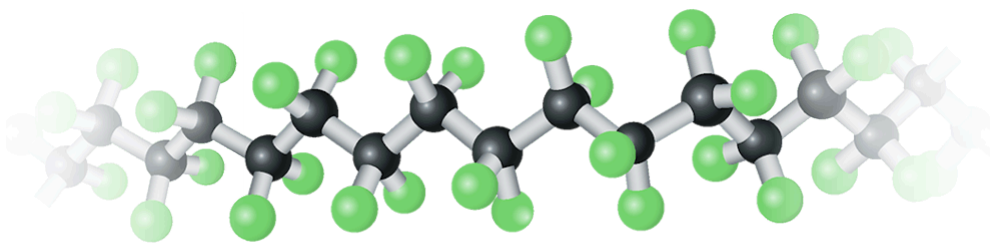
TeflonTM is a material commonly used in non-stick cookware. You might have a frying pan in your kitchen that has a teflon coating on the cooking surface. Teflon is made of the polymer tetrafluoropolyethene (PTFE). This polymer has a relatively high melting point (327°C), it does not interact with water and is very unreactive with most substances. These properties make it ideal for use in cookware. The structure of tetrafluoroethene is shown in **Figure 1**.



 More information

The image is a structural diagram of the chemical compound tetrafluoroethene (TFE), which is a monomer used to produce the polymer polytetrafluoroethene (PTFE, commonly known as TeflonTM). The diagram shows two carbon atoms (C) double-bonded to each other at the center. Each carbon atom is also bonded to two fluorine atoms (F), positioned at the top and bottom. This configuration creates a linear, symmetrical structure. The chemical formula is often represented as C₂F₄. TFE is a colorless gas under standard conditions, and the inclusion of fluorine atoms makes it highly resistant to chemical reactions, contributing to its non-stick properties.

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Ball and stick model

Figure 1. Structure of polytetrafluoroethene. More information for figure 1

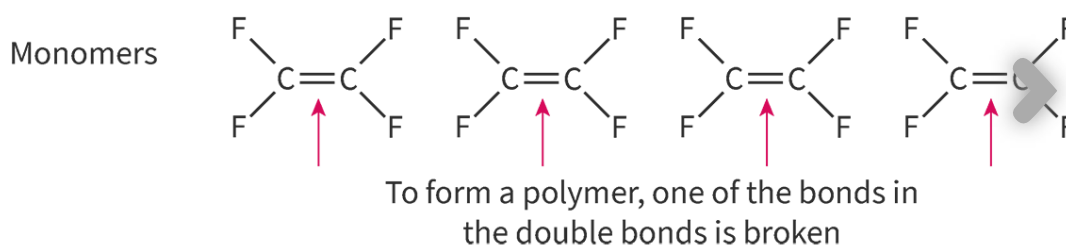
The image is a diagram illustrating the molecular structure of polytetrafluoroethene (PTFE). It depicts a repeating chain of carbon atoms, each bonded to two fluorine atoms. The diagram is arranged linearly, indicating a straight chain polymer. The carbon atoms appear in black, while the fluorine atoms are shown in green. The bonds between the atoms are represented by lines connecting individual spheres, which signify atoms. This diagram helps visualize the molecular construction of PTFE, emphasizing the alternating pattern and linear structure typical of this polymer.

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But how is a polymer such as PTFE formed? What is the structure of the monomer molecule building blocks used to form PTFE? What happens to the monomer in order to form a polymer like this?

Recall from [section S2.4.4 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/polymers-id-43845/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/polymers-id-43845/) that polymers are large macromolecules made from repeating subunits called monomers. These monomers act as the building blocks to form polymers. One method used to form a polymer is through a process called addition polymerisation. Addition polymerisation is the process of forming a molecule by simply adding monomers together. For this to occur, the monomer must have a double bond between two carbon atoms in its structure. When the polymer is formed, one of the bonds in the double bond is broken in each monomer allowing a single bond to form with other monomers. This process is illustrated in **Interactive 1** for the formation of the polymer polytetrafluoroethene. Press the arrow to view the slides.

Formation of polytetrafluoroethene



Interactive 1. Formation of Polytetrafluoroethene.

 More information for interactive 1

A slideshow with four slides displays the formation of polytetrafluoroethene.

Slide 1: Each monomer consists of two carbon atoms double-bonded to each other, with each carbon atom bonded to two fluorine atoms. Red arrows point toward the double bonds between carbon atoms to highlight the breaking of one bond. Onscreen text: To form a polymer, one of the bonds in double bonds is broken.

Slide 2: Each monomer now consists of one single bond between carbon atoms, with each carbon bonded to two fluorine atoms. Arrows from the carbon atom of one monomer to the carbon atom of the next monomer are drawn. The onscreen text below reads as follows: These carbon atoms are not stable. These monomers bond with adjacent monomers to become more stable.

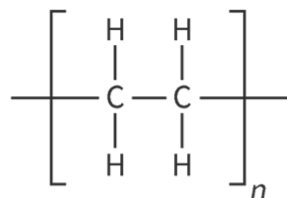
Slide 3: The image demonstrates the polymerisation process where the double bond between carbon atoms is broken, forming a continuous chain of carbon atoms with fluorine atoms attached. Onscreen text: These newly formed bonds between monomers form a polymer. Since monomers were simply added together these polymers are called addition polymers.

Slide 4: The polymerisation process of tetrafluoroethene (C_2F_4) to form polytetrafluoroethene. On the left side, the monomer structure shows two carbon atoms double-bonded to each other, with each carbon atom individually bonded to two fluorine atoms. The notation n indicates multiple monomer units. An arrow points to the polymer structure on the right side, where the repeating unit consists of single bonds between carbon atoms, each bonded to two fluorine atoms. The polymer chain is enclosed in brackets with the n notation on the outside, signifying numerous repeating units.

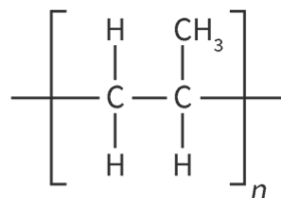
The interactive demonstrates the polymerisation reaction of tetrafluoroethene.

Notice in **Interactive 1** how the polymer formed is named after the monomer with the prefix poly- added in front. This is how all addition polymers are named. Note that many of the plastics we looked at in [section S2.4.4](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/polymers-id-43845/>) were examples of addition polymers. Looking at the plastic addition polymers shown in **Figure 2**, see whether you can name and draw the structures for each of their monomers.

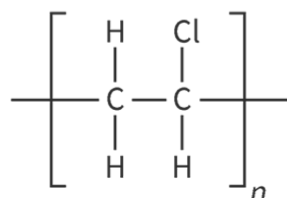
Polyethene/polyethylene



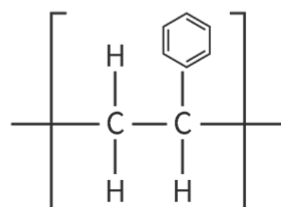
Polypropene/ polypropylene



Polychloroethene/polyvinyl chloride



Poly(1-phenylethene)/polystyrene

**Figure 2.** Common plastic addition polymers.

 More information for figure 2

This diagram presents the chemical structures and names of four common plastic addition polymers. The polymers are displayed in a grid-like format. From top to bottom, left to right: the first is Polyethylene, depicted with a chain of repeating ethylene units (CH₂-CH₂); next is Polypropylene, shown with repeating propylene units (CH₂-CH(CH₃)); followed by Polyvinyl Chloride, represented with repeating vinyl chloride units (CH₂-CHCl); and lastly, Polystyrene, shown with a chain of styrene units (CH₂-CH(C₆H₅)). Each polymer's structural formula is labeled accordingly, highlighting the differences in their monomer units. This demonstrates the variety in structure among addition polymers.

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Nature of Science

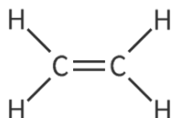
Aspect: Science as a shared endeavour

To name compounds a set of rules is set by the International Union of Pure and Applied Chemistry (IUPAC  (<https://iupac.org/>)). These rules allow scientists all over the world to communicate information about the substances they are studying. You may notice however that there is some variation in name for the same polymer labelled in commercial products. For example, polytetrafluoroethene would be the IUPAC polymer name but polytetrafluoroethylene is used commercially. Both refer to the same polymer.

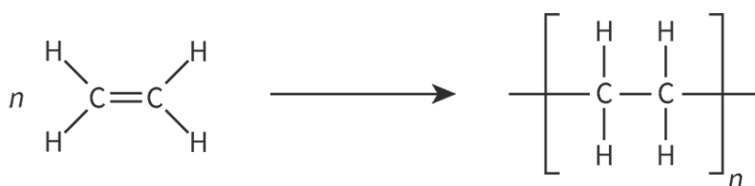
Worked example 1

What is the name and the structure of the addition polymer formed from ethene, C_2H_4 ?

The molecule C_2H_4 has the following structure:



When the polymer is formed, one of the bonds in the double bonds breaks to join monomers together to form the polymer with the following structure:



Since the polymer was made from the monomer ethene, the name would be polyethene.

Worked example 2

The following polymer is known as polyvinylidene dichloride. Identify the repeating unit and deduce the structure of the monomer for the following polymer.

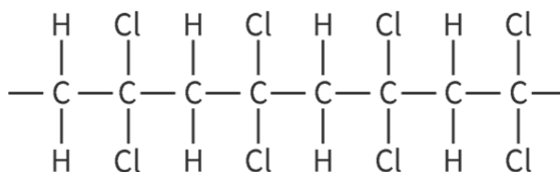


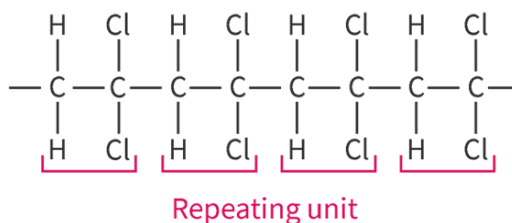
Figure 3. Structural formula of polyvinylidene dichloride.

 More information for figure 3

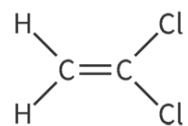
The image depicts the structural formula of polyvinylidene dichloride. It shows repeating units characterized by a vinylidene group bonded to a dichloride group. The chemical structure features a polymer backbone formed by carbon atoms, each carbon atom alternately double-bonded to another carbon atom while also sharing single bonds with two hydrogen atoms. Each repeating unit within the polymer chain includes two chlorine atoms attached to carbon atoms, contributing to the overall molecular structure of polyvinylidene dichloride. The formula is representative of how the polymer's monomer units are linked chemically to form its macromolecular configuration.

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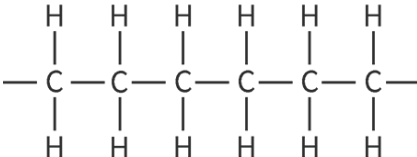
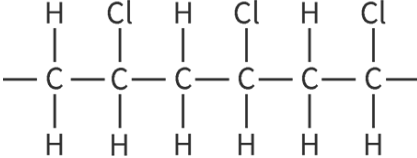
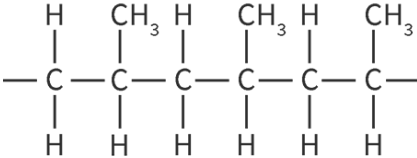
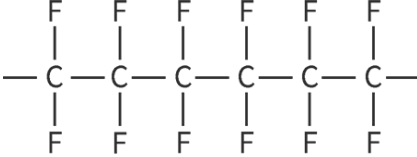
Looking at the polymer the repeating unit would be:



This would indicate that the monomer has two carbon atoms, two hydrogen atoms and two chlorine atoms in its structure. Now in order to form a polymer from the monomer a bond between carbon atoms would need to break. This means the monomer has a double bond between carbon atoms. Therefore, the structure of the monomer would be:



In **Interactive 2** drag and drop the names of monomers to match them to the correct polymer structure.

Polymer structure	Name	Monomer	
			
			
			
			
polypropene	polytetra- fluoroethene	polyethene	poly- chloroethene
tetra- fluoroethene C_2F_4	ethene C_2H_4	propene C_3H_6	chloroethene C_2H_3Cl

✓ Check

Interactive 2. Polymers and Their Monomers.

 More information for interactive 2

An interactive table illustrates polymer structures where users need to drag and drop the corresponding names and monomers for the given structures.

The provided polymer structures are described as follows:

Structure 1:

Six carbon atoms are arranged in a straight line, connected by single bonds. An extra bond extends from the first and the last carbon atoms. Each carbon atom is single-bonded to hydrogen atoms at the top and bottom.

Structure 2:

Six carbon atoms are arranged in a straight line, connected by single bonds. An extra bond extends from the first and the last carbon atoms. Each carbon atom is single-

bonded to a hydrogen atom at the bottom. At the top, hydrogen and chlorine atoms are bonded alternately to the carbon atoms.

Structure 3:

Six carbon atoms are arranged in a straight line, connected by single bonds. An extra bond extends from the first and the last carbon atoms. Each carbon atom is single-bonded to a hydrogen atom at the bottom. At the top, hydrogen atoms and methyl groups are bonded alternately to the carbon atoms.

Structure 4:

Six carbon atoms are arranged in a straight line, connected by single bonds. An extra bond extends from the first and the last carbon atoms. Each carbon atom is single-bonded to fluorine atoms at the top and bottom.

The drag-and-drop options are provided as follows:

Names:

Polypropene

Polytetrafluoroethene

Polyethene

Polychloroethene

Monomers:

Tetrafluoroethene C_2F_4

Ethene C_2H_4

Propene C_3H_6

Chloroethene C_2H_3Cl

This interactivity enables users to identify the names and monomers of various polymer structures.

Read below for the solution:

Structure 1:

Name: Polyethene

Monomer: Ethene C_2H_4

Structure 2:

Name: Polychloroethene

Monomer: Chloroethene C_2H_3Cl

Structure 3:

Name: Polypropene

Monomer: Propene C_3H_6

Structure 4:

Name: Polytetrafluoroethene

Monomer: Tetrafluoroethene C_2F_4

Creativity, activity, service

Strand: Service

Learning outcomes:

- Recognise and consider the ethics of choices and actions.
- Demonstrate engagement with issues of global significance.

Polymers are often chosen for their durability and longevity. However, it is these properties that result in plastics causing long-term problems [↗](https://www.worldwildlife.org/stories/the-problem-with-plastic-in-nature-and-what-you-can-do-to-help) (<https://www.worldwildlife.org/stories/the-problem-with-plastic-in-nature-and-what-you-can-do-to-help>) in the natural environment. According to the WWF, a plastic straw can take up to 200 years to break down.

Reducing the use of single-use plastic is one way in which we can all make a difference. Another way is to raise awareness of the importance of reducing plastic waste. Read about more ways to reduce the use of plastics in your school here [↗](https://www.sustainablemerton.org/post/what-can-schools-do-to-reduce-plastic-pollution) (<https://www.sustainablemerton.org/post/what-can-schools-do-to-reduce-plastic-pollution>).

The next activity will allow you to identify the names and structures of varying addition polymers and/or their monomers, along with researching the uses for each polymer.

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:** Research skills — Using search engines and libraries effectively
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Individual activity

Complete the following worksheet.

Worksheet [↗](https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Chemistry)
(https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Chemistry)

In this worksheet you will need to complete the following:

- Deduce the name and structure of a polymer given the monomer.
- Deduce the name and structure of the monomer given the polymer.
- Research any practical applications and uses each polymer has.
- Check your worksheet answers with a classmate or when you research the practical uses of polymers.